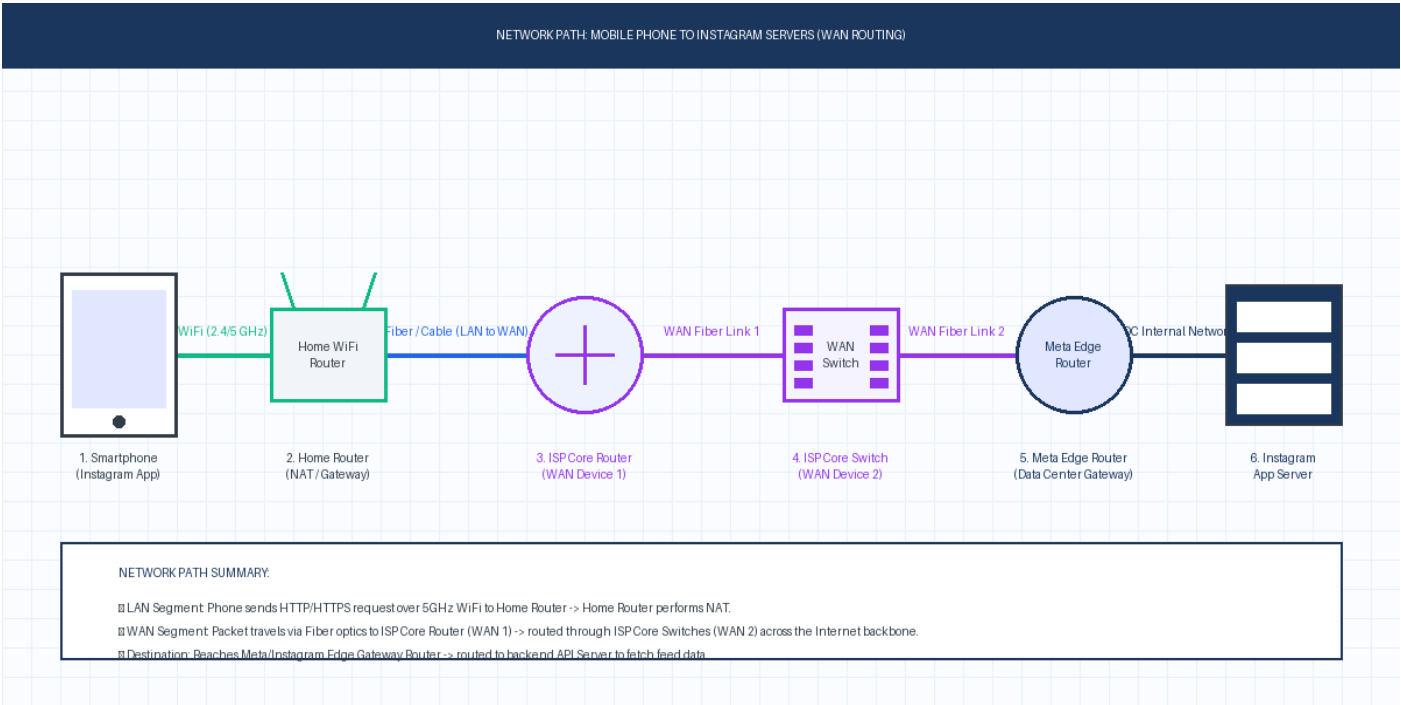


TASK-1



TASK-2

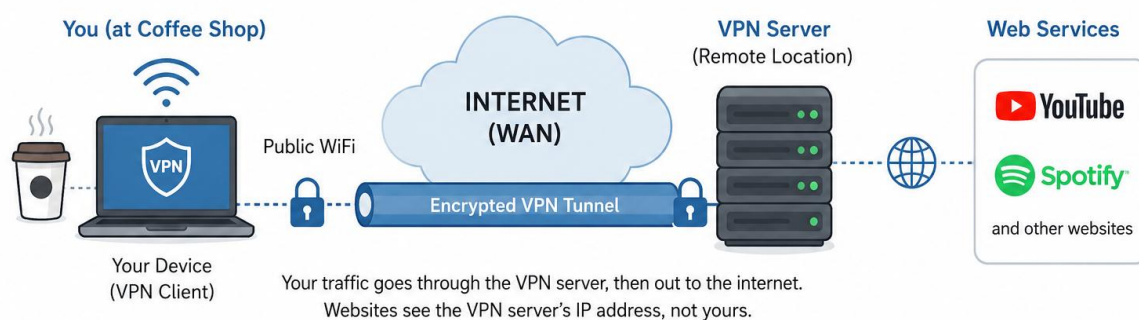
WAN Technology	Typical Speed	Relative Cost	Real-World Application Example
Leased Line (<i>Dedicated Point-to-Point</i>)	Fixed & Symmetric (e.g., 64 Kbps up to 10 Gbps with guaranteed 1:1 bandwidth)	Very High (High recurring monthly fixed fee regardless of usage)	IRCTC Ticketing & Banking Core: Used for critical, uninterrupted connections between Indian Railways ticketing hubs/banks and core datacenters where guaranteed uptime and low latency are non-negotiable.
MPLS (<i>Multi-Protocol Label Switching</i>)	Flexible & Scalable	High (More cost-effective than)	Zomato Corporate Offices & Regional Warehouses: Used to connect regional corporate branch offices, inventory hubs, and enterprise systems with prioritized voice, video, and database traffic across cities.

WAN Technology	Typical Speed	Relative Cost	Real-World Application Example
	(Typically 2 Mbps to 1 Gbps with Traffic Prioritization/QoS)	private Leased Lines, but pricier than Broadband)	
Broadband Internet (FTTH / Fiber / DSL)	Asymmetric High-Speed (e.g., 50 Mbps up to 1 Gbps download; shared contention ratio)	Low (Budget-friendly, highly accessible fixed-line plans)	Your Home & Local Retail Outlets: Used for everyday home Internet access (streaming Instagram, browsing, YouTube), as well as small local restaurants/stores to process UPI payments or view orders.

TASK-3

How VPN Uses WAN to Secure Your Connection

A VPN (Virtual Private Network) uses the public WAN (the Internet) to create a secure, encrypted tunnel between your device and a VPN server. This tunnel hides your real IP address, encrypts your data, and protects your online activity from hackers, ISPs, and other users on the same public WiFi.



How it works:

1. You connect to a VPN server over the public Internet (WAN) using an encrypted tunnel.
2. All your internet traffic (e.g., Spotify, YouTube) goes through this tunnel.
3. The VPN server forwards your requests to the websites.
4. Websites send the data back to the VPN server, which sends it securely to you.
5. Your real IP address is hidden, and your data is encrypted—keeping you safe on public WiFi.

TASK-4

For Flipkart to connect its major fulfillment centers and warehouses across **Ahmedabad, Mumbai, and Delhi**, the best WAN technology choice would be **MPLS (Multi-Protocol Label Switching)**, combined with a **Secondary / Hybrid SD-WAN (Software-Defined WAN)** backup over enterprise broadband for redundancy.

Why MPLS is the Best Choice for Flipkart

1. Low Latency & High Reliability for Real-Time Inventory & Order Syncing

Flipkart's warehouse operations rely heavily on real-time data processing—such as scanning barcodes during order packing, updating live inventory stock counts across India, and synchronizing with delivery dispatch servers.

- **MPLS** provides dedicated, private routing pathways with strict **Service Level Agreements (SLAs)** guaranteeing 99.99% uptime and fixed low latency. Unlike public broadband, data packets between Ahmedabad, Mumbai, and Delhi won't suffer from congestion, packet loss, or unpredictable lag during major festive sale events (like *Big Billion Days*).

2. Built-in Traffic Prioritization (Quality of Service - QoS)

An enterprise warehouse network handles many types of traffic simultaneously: critical database calls (inventory scans), VoIP/video communication between warehouse staff and corporate offices, and routine log uploads.

- **MPLS** allows Flipkart to tag and prioritize mission-critical data packets (like live order processing) over less sensitive traffic (like daily backup uploads). This ensures that core operational tasks remain fast and uninterrupted even during peak bandwidth spikes.